

BOEKBESPREKINGS/BOOK REVIEWS

Combination of one-sided statistical tests by J. Oosterhoff

First edition, 148 pages, 11 figures and 4 tables.

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Every practising statistician is often confronted with the problem of combination of statistical procedure, either of the test statistics themselves or of the outcomes (i.e. exceedance probabilities) of the statistical tests based on these test statistics. Although quite a number of papers on combination procedures have appeared, only in a few papers the theoretical properties of combination procedures (such as power or performance under alternative hypotheses) have been considered.

The present book is the first in this field and, being a continuation of a paper by W.R. van Zwet and the author on decision theoretical aspects of the one-sided combination problem published in 1967 in the *Annals of Mathematical Statistics*, it extends and generalises these results. After formulating the general combination problem, the first chapter gives a historical survey on these procedures, followed by a short treatment of basic decision theoretical concepts and theorems. The main concern of this book is centered on most stringent tests and the concept of shortcoming of a test (which is defined as the amount by which the power of the test falls short of the envelope power) plays an important rôle in the development of the theory.

In the second chapter, after reducing the one-sided combination problem to a problem of combining normally distributed random variables with known variances by an asymptotic approach, the most stringent combination procedure is obtained for this standard problem in a few cases. It is also shown that the likelihood ratio combination procedure and Fisher's omnibus test (combination of one-sided exceedance probabilities of independent tests) are asymptotically optimal in some sense for this problem as the significance level tends to zero. Finally, several combination procedures are compared numerically.

In the third chapter the combination of Student t-tests is considered. For invariant procedures some results are derived in the case of equal variances, but not much progress has been made in the case of unequal variances.

In the final chapter the L.R. test in multinomial distributions is studied and the relevance of this study to combination problems is briefly outlined.

This work, which in a certain sense can be considered as a research report, is a very valuable contribution to the theory of combination procedures, but is by no means easy to read. It contains relatively few printing errors.

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